

50

System Design

Interview Questions

*Seen in **ML Engineer** and **AI Engineer** interviews
at FAANGs, startups and consulting firms*



Core Concepts

1. **Scalability:** Understand horizontal vs. vertical scaling, and how to design scalable systems.
2. **Load Balancing:** Concepts behind tools like Nginx, HAProxy.
3. **Caching:** Mechanisms (e.g., Redis, Memcached) and strategies (LRU cache, write-through, write-back).
4. **Databases:** Differences between SQL (e.g., PostgreSQL, MySQL) and NoSQL (e.g., MongoDB, Cassandra). ACID vs. BASE properties.
5. **Consistency vs. Availability:** CAP theorem.
6. **Distributed Systems:** Challenges (e.g., network failures, partition tolerance) and patterns (e.g., leader election).
7. **Message Queues:** Usage and tools like RabbitMQ, Kafka.
8. **Microservices:** Benefits, challenges, and communication methods (e.g., REST, gRPC).
9. **API Design:** RESTful services, rate limiting, versioning.
10. **Replication & Sharding:** Strategies to distribute data.
11. **Failover and Recovery:** Strategies to handle system failures.
12. **Monitoring & Logging:** Tools and strategies for maintaining system health.

Interview Questions

1. Design a parking lot management system.
2. Design an API rate limiter.
3. How would you handle security concerns in a distributed system?
4. Design a system to handle millions of events per second.
5. Design an online booking system like Airbnb.
6. How would you design for high availability in a critical application?
7. Design a photo-sharing service like Instagram.
8. Discuss the CAP theorem and its implications for system design.
9. Design a system for processing big data in real-time.
10. How would you design a distributed file storage system like Google's GFS or Hadoop's HDFS?
11. Design an ad-serving platform.
12. Explain strategies to detect and prevent fraud in an online transaction system.
13. Design a system for real-time analytics.
14. How would you design a trending topics feature for a platform like Twitter?
15. Design an email sending service.
16. How would you ensure data consistency in a microservices architecture?
17. Design a calendar system.
18. Discuss strategies for zero-downtime deployments in critical systems.
19. Design a system for tracking user actions on a website.
20. How would you optimize for read-heavy vs write-heavy systems?

Interview Questions

21. Design a URL shortening service like Bitly.
22. Design a system for autocomplete functionality.
23. Design a distributed caching system.
24. How would you design a global content delivery network (CDN)?
25. Design a rate limiter for APIs.
26. How would you design a logging and monitoring system?
27. Design an online multiplayer game backend.
28. How would you handle versioning in a microservices architecture?
29. Design a system to efficiently find 10 top selling products on an online marketplace.
30. Explain how you'd handle data replication and consistency across distributed databases.
31. Design a recommendation system like the ones on Amazon or Netflix.
32. How would you handle schema migrations in a large-scale database system?
33. Design a notification service for a real-time application.
34. How would you design a chat application like WhatsApp or Slack?
35. Design a search engine.
36. How would you design the backend for a social network like Facebook or Twitter?
37. Design a distributed message queue system.
38. Explain the design behind a load balancer.
39. Design a key-value store.
40. How would you handle hotkeys in a caching system?

Interview Questions

41. Design a real-time traffic reporting system.
42. Discuss database sharding strategies.
43. Design an e-commerce checkout system.
44. How would you handle a system failure in the middle of a transaction?
45. Design an order management system for an e-commerce platform.
46. Discuss strategies for rolling back a faulty deployment.
47. Design a ride-sharing service like Uber or Lyft.
48. How would you design a system to support multiple languages and regions?
49. Design a web crawler.
50. How would you handle database connection pooling in a large-scale application?

Join DataInterview.com

Access courses, mock interviewing and private community with coaches and candidates like you



Created by interviewers at top companies like Google and Meta

